SARA GRAÇA

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PROFILE

Master's student in Computational Linguistics with research experience in NLP. Proficient in Python, with hands-on experience applying AI, including ASR model fine-tuning, building RAG pipelines, and training transformers. Multilingual profile (English, French, Portuguese, Japanese). Seeking a final-year Master's internship (M2).

EDUCATION

M1-M2 Computational Linguistics *2025 - 2027*
Université Paris Cité | M1 Grade: 17.6/20
Key courses: RAG & AI agents; Theory & practice of LLMs; Machine learning for NLP; Graph algorithms

Bachelor's in Translation *2022 - 2025*
Portuguese/English/French | NOVA FCSH | Lisbon | Grade: 19/20
Received merit scholarship
Organised conferences

Bachelor's in Applied Mathematics and Computation (partial completion) *2020 - 2022*
Instituto Superior Técnico | Lisbon

PROFESSIONAL EXPERIENCE

Research Internship *May 2026 - July 2026*
LACITO, CNRS | Paris | 2.5 months

- Improved an existing full ASR pipeline, designed to be extensible to new languages and producing multi-format output (.tsv/.xml/.eaf) for downstream tools
- Fine-tuned `wav2vec2-large-xlsr-53` on low-resource languages Thulung (13h) and Japhug (57h), achieving WERs of 45.3% and 15.7% respectively, with optional KenLM beam-search decoding
- Designed Python scripts for conversion between annotation formats ELAN (`.eaf`) and Pangloss (`.xml`)

NLP Annotator *October 2026*
Annotation of data for NLP at the University of Helsinki

Private Tutor *2020 - 2025*
Taught mathematics, physics-chemistry, and geometry at high school level

PROJECTS

Audio Deepfake Detection

Benchmarked 6 feature sets on speaker-disjoint Portuguese TTS speech: deep representations (XLS-R, Whisper) reached AUC 0.994 vs 0.69–0.78 for handcrafted ones (MFCC, CQCC...), with significance testing and acoustic error analysis.

RAG System and Chatbot

Developed an end-to-end RAG system: PDF parsing, retrievers for both text and SQL data, a chatbot, and an evaluation pipeline to measure retrieval and answer quality.

Language-Specific Neurons in mBERT

Reimplemented Tang et al. (2024) on mBERT: ablating each language's neurons raised its own masked-LM loss at least 3× more than random ablation.

SKILLS

Programming & Tools: Python, Git, DVC, pixi
ML & NLP: PyTorch, Hugging Face Transformers, spaCy, scikit-learn, KenLM
Speech & Audio: librosa, soundfile, torchaudio
LLMs & Retrieval: RAG, embeddings, LLM evaluation

Languages: Portuguese (native) | English (C2) | French (C1) | Japanese (A2)